3 A projectile of mass 2.0 kg is launched on the Earth with some initial velocity. Another projectile of mass 4.0 kg is launched on the Moon with the same initial velocity. The acceleration of free fall on the Moon is 1.6 m s^{-2}.

Neglecting air resistance, what is the ratio $\frac{\text{range of projectile on the Earth}}{\text{range of projectile on the Moon}}$?

A 0.16 **B** 0.33 **C** 3.3 **D** 6.1